Alireza Kamali Ardakani

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Education

BS **University of Tehran**, Mechanical Engineering — Sep 2020 – Sep 2025
GPA: 3.83/4 (18.04/20) — Ranked 9th/120
Thesis: *Dynamic Analysis of Gough–Stewart Parallel Manipulators in a Serial Configuration*

SAMPAD High Schools, Mathematics — Aug 2017 – Aug 2020
GPA: 4.0/4.0 (19.64/20) — Ranked 1st/30

Research Interests

- Robotic Manipulation
- Dynamics and Control Systems
- Mechatronics & Sensors
- Autonomous Systems
- Legged Locomotion
- Physics-Informed AI
- Biomechanics & Rehabilitation
- Haptic and Wearable Devices
- Bio-Inspired Mechanisms

Publications

Published

- A. Kamali et al., “Dynamic Modeling of Double Segment Redundant Gough–Stewart Hybrid Manipulator,” *IEEE ICROM 2024*.
- S. Rahmati, A. Kamali et al., “Design, Fabrication, and Control of a Mechanical Integrator,” *ISME 2025*.

Manuscript in Preparation

- H. Akbari, A. Kamali, et al., "Dynamic Analysis and Base Inertial Parameters of Gough–Stewart Hyper-Redundant Manipulator."

Experience

Research Assistant, University of Tehran — Human–Robot Interaction Laboratory — Apr 2024 – Present
Supervisor: Dr. Mehdi Tale Masouleh

- Dynamic modeling and control of serial–parallel and hyper-redundant manipulators.
- Redundancy resolution, obstacle avoidance, and minimal inertial parameter identification.
- Simulation and validation using MATLAB/Simscape, Python, and MSC Adams.

Research Assistant, University of Tehran — Vibration Nonlinear Systems Laboratory — Jan 2026 – Present
Supervisor: Dr. Arash Bahrami

- Design, fabrication, and control of a soft Fin-Ray robotic gripper.
- Vision-based grasping of deformable objects with emphasis on stable and compliant manipulation.
- SolidWorks modeling, computer vision, system identification, and machine learning.

Teaching Assistant, University of Tehran

Spring 2026	Mechanical Vibrations	Dr. Bahrami
Fall 2025	Physics II	—
Fall 2024	Robotics (Graduate)	Dr. Tale Masouleh
Fall 2024	Product Design and Development	Dr. Hamedi
Spring 2024	Measurement Systems	Dr. A. Shirazi
Fall 2023	Mechanical Vibrations	Dr. Bahrami
Spring 2023	Numerical Methods	—
Fall 2022	Mechanics of Materials I	Dr. Mahnama

Projects

Detailed implementations and code: github.com/alirezakamali80

Reinforcement Learning Control of a One-Link Robotic Arm 2026
Built a custom Gymnasium-style robotic-arm simulator and implemented SARSA, Q-learning, DQN, and SAC controllers for swing-up and stabilization. Evaluated performance using success rate, RMS angle error, control effort, and closed-loop trajectories.

ATRA: Vision-Guided SCARA Robotic Arm 2026
Contributed to a three-axis SCARA robotic arm for automated pick-and-place and sorting. Integrated YOLOv8 and HSV-based object detection with a compliant Fin-Ray soft gripper for safe handling of delicate and irregular objects.

Advanced Visual Perception with ViT and YOLO 2026
Fine-tuned ViT-B/16 and ResNet-50 for driving-scene classification on BDD100K, and trained YOLOv11 for UAV object detection on VisDrone. Evaluated models using ROC/AUC, mAP, precision/recall, confusion matrices, and inference speed.

Satellite Attitude Control System Design 2026
Modeled and discretized a flexible satellite attitude-control system in MATLAB. Designed a notch-filtered double-lead compensator and compared Tustin, matched pole-zero, and ZOH digital implementations.

Dynamic Modeling of Serial Gough–Stewart Robots 2024
Developed inverse-dynamics models for serial–parallel Gough–Stewart manipulators using virtual work and Jacobian-based formulations. Implemented MATLAB/Simscape simulations for trajectory tracking, inverse kinematics, and actuator-force validation.

Mechanical Integrator 2024
Designed and fabricated a stepper-motor-driven mechanical integrator. Modeled the lead-screw mechanism and experimentally validated integration through output-shaft displacement measurements.

Robotics Mini Projects 2024
Implemented robotics projects involving manipulator kinematics and dynamics, SCARA modeling, motion sensing, ROS simulation, and deep-learning-based perception.

Solenoid-Driven Rotary Motor with Speed Control 2023
Designed and fabricated a solenoid-driven rotary motor. Developed dynamic models and implemented PI closed-loop speed control with simulation and experimental validation.

Skills

Programming: Python, C/C++ **Software:** MATLAB, Simulink/Simscape, SolidWorks, AutoCAD, ROS, Linux **Simulation:** MSC Adams, COMSOL, Proteus **AI:** Machine Learning, Neural Networks, Deep Learning, Computer Vision **Miscellaneous:** Arduino, LaTeX

Selected Courses

Artificial Intelligence	4	Advanced Control	4
Robotics	4	Mechatronics	4
Automatic Control	4	Dynamics	4
Programming	4	Electronics	4
Measurement Systems	4	Engineering Mathematics	4
MEMS/NEMS	4	Biomechanics	4

Language and Test Scores

- **Persian:** Native
- **English:** Professional Working Proficiency

IELTS 6.5 Nov 2024

Honors and Awards

- Among the Top 15 finalists of the National Mechanical Engineering Student Olympiad (2023, 2024).
- Ranked among the top 0.2% in the National University Entrance Examination (2020).

Voluntary Activities

Conference Organizer — ICROM 2023 Dec 2023
Editor & Author — Gaj International Publications, Chemistry Sep 2020 – Sep 2021
Chemistry Teacher — Self-employed Sep 2020 – Mar 2022

References

Dr. Mehdi Tale Masouleh Associate Professor, Human-Robot Interaction Laboratory, University of Tehran
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Dr. Arash Bahrami Associate Professor, University of Tehran
arash.bahrami@ut.ac.ir